

Structured Semantic Components

$p = (e, c, Q, d, o)$



Event

Market pulse



Context

Regime window /
reference condition



Qualities

Checklist / filter



Direction

Continuation-reversal



Output

Signal target /
rank score

$p = (e, c, Q, d, o)$

Each trading hypothesis is decomposed
into interpretable semantic fields

Search & Selection

Generate Schema Plans

Semantic Space



EXAMPLE CASE

Event	Price Breakout
Context	Recent High Region
Qualities	Volume Confirmation + Multi-horizon Consistency
Direction	Continuation
Output	Cross-sectional Rank

Plan 1:
E1 + C3 + Q2 + D1 + O2

Plan 2:
E2 + C1 + Q4 + D2 + O3

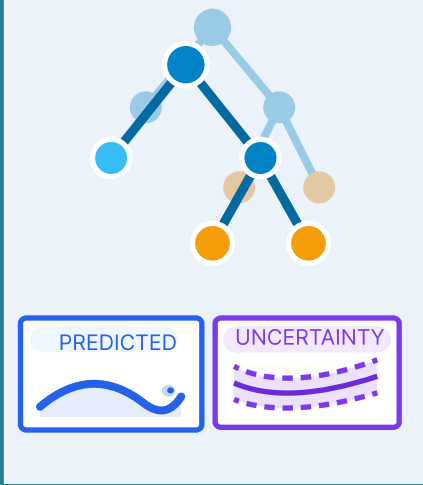
Plan 3:
E5 + C2 + Q1 + D1 + O1

⋮

Plan N:
Ei + Cj + Qk + Dm + On

Quota Scheduler

LightGBM
Ensemble



Quota Selector

High Reward

Uncertainty

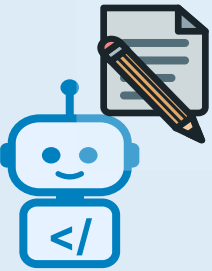
Novelty

Local Mutation

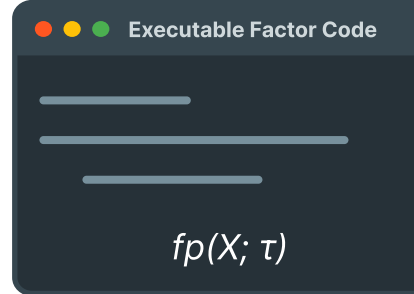
Plan

Plan-to-Code Realization

LLM Agent:
Realize Factors



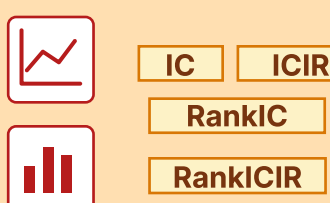
Executable Factor Code



Multi-layer Validation & Backtest

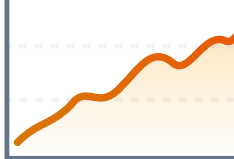
- ✓ Contract Check
- ✓ Numerical Validity
- ✓ Leakage Check
- ✓ Backtest Harness
- ✓ Backtest Reward

Objective Metrics



CSI300 Portfolio
Evaluation

Validated
Factor Pool



Reward Buffer

Schema Plan

Reward

Schema-level Semantic
Feedback